UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 4 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION 1 (5 marks)

Give suitable example for each processor organization.

QUESTION 2 (5 marks)

a) List TWO (2) advantages of SMP organization.
b) Define cluster organization.
c) Give ONE (1) benefit of cluster organization.

QUESTION 3 (5 marks)

a) Name the inventor of the integrated circuit.
b) In which computer generation was the integrated circuit first introduced?
c) List THREE (3) main components of an integrated circuit.

QUESTION 4 (5 marks)

Differentiate between DRAM and SRAM in term of the following characteristics:

a) Speed.
b) Size per bit.
c) Cost.
d) Type of bits stored.
e) Refreshing need when powered.

QUESTION 5 (10 marks)

a) Compare TWO (2) system bus timing approach with appropriate examples.
b) Discuss ONE (1) reason why an I/O module is needed.
c) List TWO (2) interfaces to external devices

QUESTION 6 (10 marks)

a) Explain how the bits are stored in the SRAM and DFiAM.
b) Why is SRAM technology faster than DRAM technology?
c) Compare SRAM and DRAM in terms of cost and power consumption.

QUESTION 7 (10 marks)

a) Convert 273.7812510 to a floating-point number in IEEE single precision format in

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hexadecimal value form.

b) Convert C158 8000H in IEEE single precision floating point number to decimal value form

QUESTION 8 (10 marks)

Keyboard is the most common input/output device to communicate between computer and user.

a) What is the input/output module category for keyboard?

b) State TWO (2) other devices that are in the similar input/output module category?

c) What is the basic unit of exchange for keyboard?

d) Discuss TWO (2) types of characters.

QUESTION 9 (10 marks)

Describe FOUR (4) type of access methods based on hardware implementation of the storage device. Provide ONE (1) appropriate example for each access type.

QUESTION 10 (10 marks)

a) Differentiate SRAM and DRAM from the perspective of the following characteristics,

i. The way the bits are stored.

ii. Speed.

iii. Location it is being used.

b) Why does DRAM need refreshing?

c) Comparing DRAM and SRAM, which memory would be placed higher in the memory hierarchy? Give your justification to support your answer.

PART B (20 MARKS)

Answer the question in this part.

QUESTION 11 (20 marks)

Question 1 is based on Figure 1 and Table 1.
Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 1 using one-, two- and three-address instructions. The instructions available for use are provided in Table 1. Note that Register A, B, C, D, E and F cannot be modified and Register X, Y and Z can be used as temporary register. Final answer must be stored in Register X.

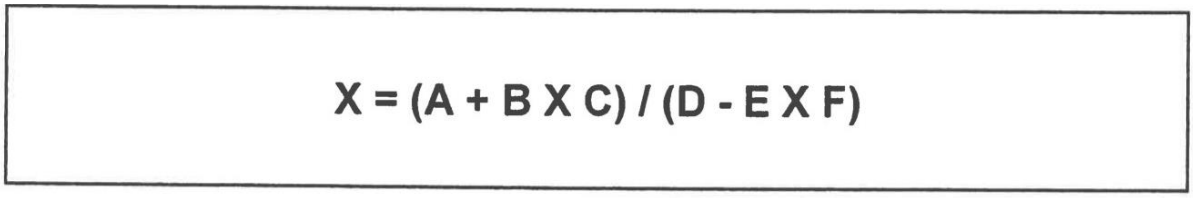


Figure 1

Table 1 : Machine Instruction

i Nia dress		**address**
LOAD M	MOVE (X e Y)	MOVE (X e Y)
STORE M	ADD (X k.. X + Y)	ADD (X e Y + Z)
ADD M	SUB (X e X - Y)	SUB (X e Y-Z)
SUB M	MUL (X e X x Y)	MUL (X e Y x Z)
MUL M	DIV (X e x / Y)	DIV (X e Y / Z)
DIV M		

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